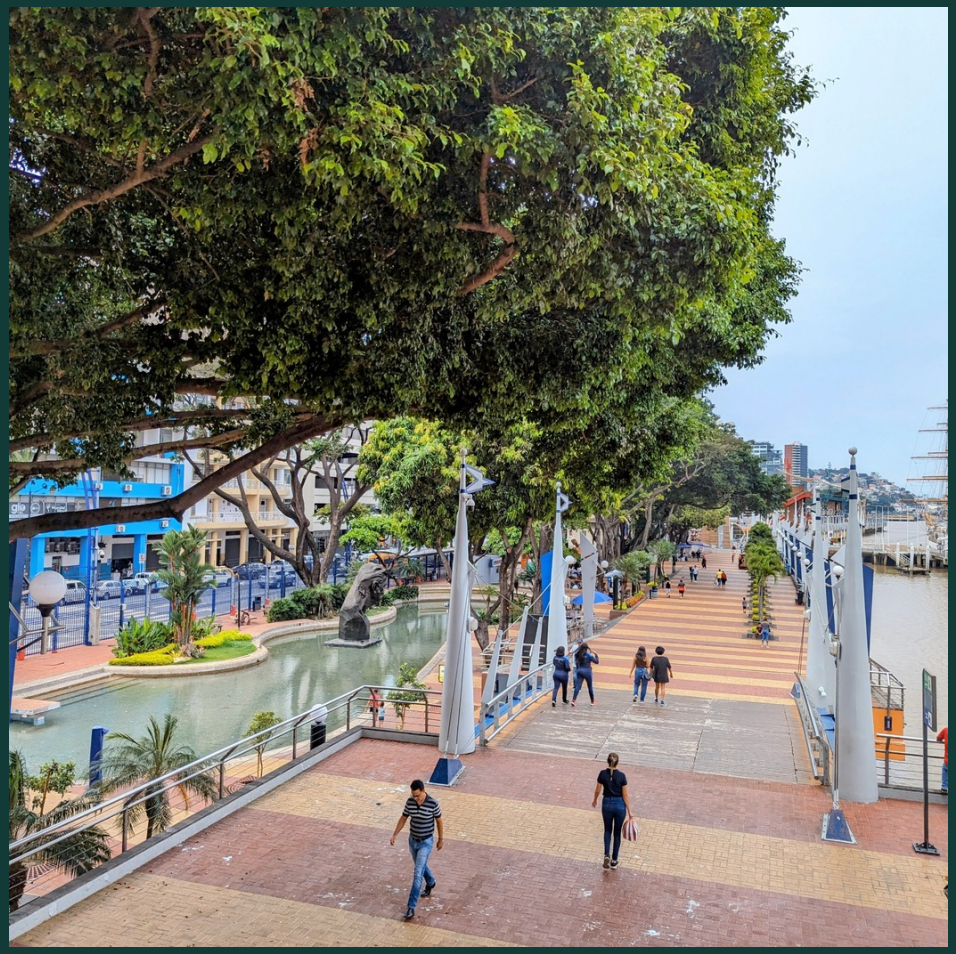
PROJECT PROPOSAL

FLOODSHADOW

Connecting forecasts, urban infrastructure and rainfall impacts

A rainfall decision workflow for Guayaquil: anticipation, monitoring and evidence-led follow-up in one place.

BEFORE → DURING → AFTER



Cover context photograph: Guayaquil waterfront · Freddy eduardo · CC BY-SA 4.0 · Cropped.

TEAM 3JOTA

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Julinho Jesús Plua Mendosa

26 October 2026 · English submission

01 / THE URBAN PROBLEM

Rain affects how the city works

Guayaquil needs to connect forecasts, infrastructure and documented impacts to anticipate problems and respond better to intense rainfall.

The information is fragmented

A weather outlook says what may arrive. Territorial information shows what is exposed. Incident records describe what people experienced. These pieces need a shared place and time.

The consequences extend beyond flooding

Water accumulation can interrupt street access, strain drainage and affect journeys to schools, health care or other essential services. The useful question is where attention is needed and why.



Conceptual illustration · Rainfall, street access, drainage and essential services.

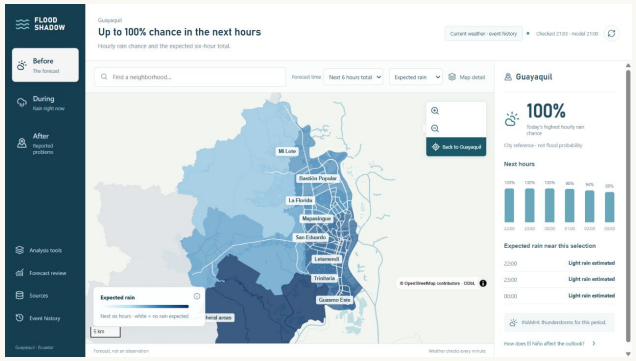
Project focus: support local analysts, community coordinators and verification teams in Guayaquil with a connected geographic view.

02 / THE COMPLETE SOLUTION

One event through three connected stages

The same area remains in focus while the application moves from the outlook to changing conditions and then to evidence and follow-up.

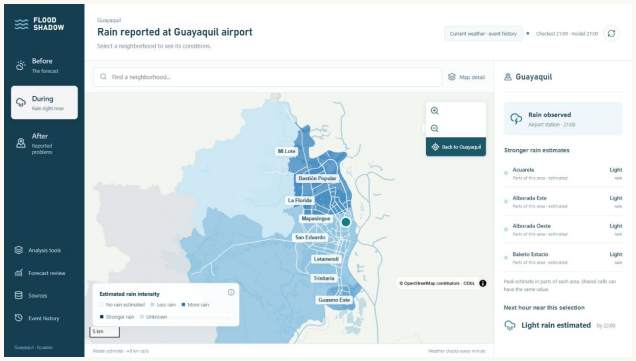
BEFORE



Anticipate problems

Inspect the next six-hour outlook and identify places that deserve preventive attention.

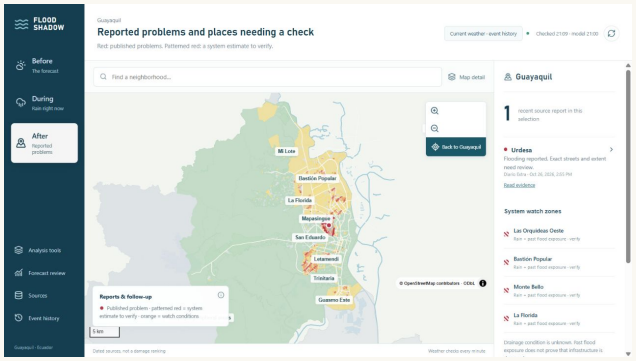
DURING



Monitor rainfall

Follow the latest estimates, source times and area summaries as weather information changes.

AFTER



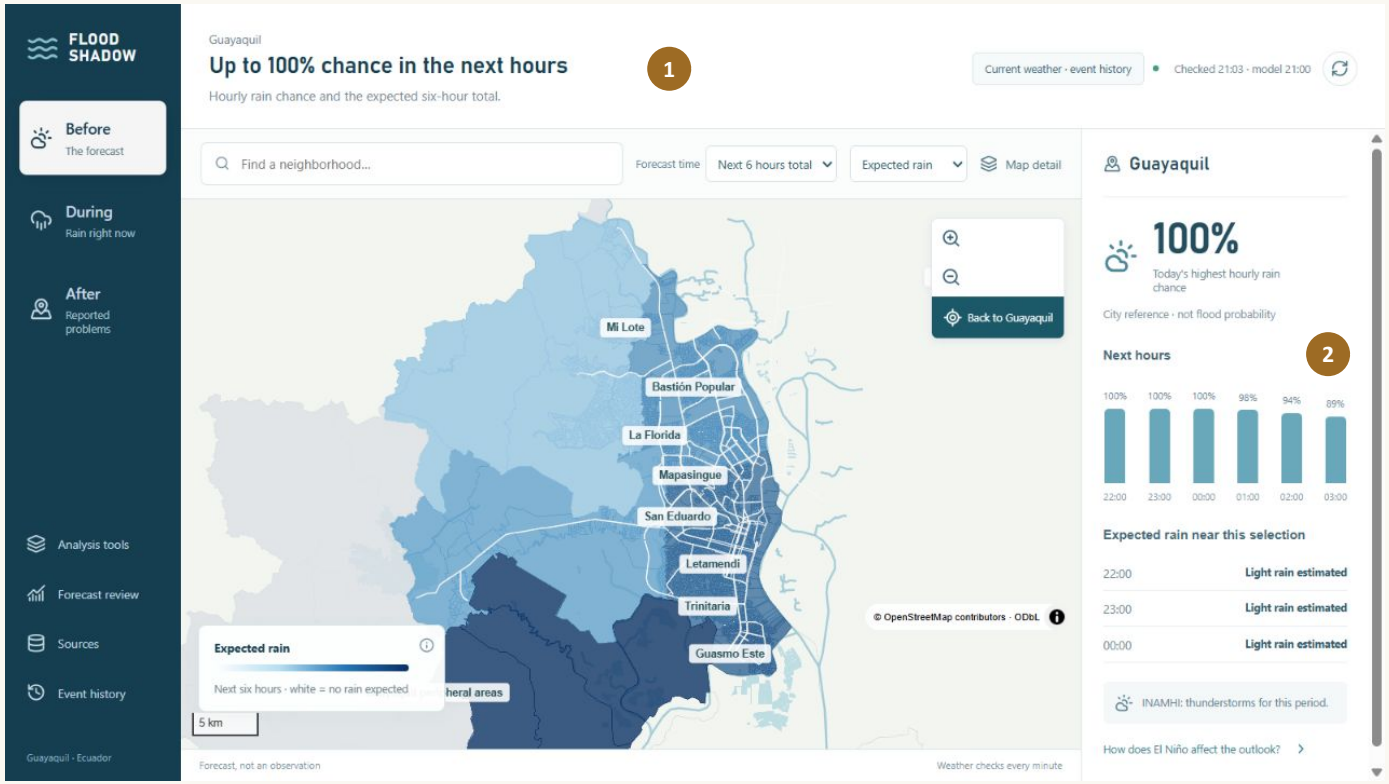
Review impacts and learn

Read documented impacts, identify verification priorities and preserve what the event teaches.

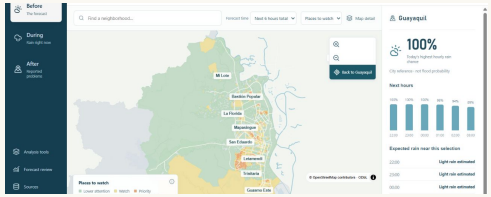
Verified outcomes feed future assessments: retain the event, review the earlier outlook and improve the next evaluation.

03 / BEFORE

See the outlook before the problem develops



Real beta captures · 26 October 2026 · The percentages shown describe rain likelihood, not the probability of flooding.



1 Choose a time

The default view combines the next six available forecast hours. An individual hour can be selected to understand when rain is expected.

2 Read the forecast

Probability bars and rainfall descriptions make the outlook understandable without meteorological expertise.

Preventive attention: the watch layer combines forecast rainfall with mapped territorial exposure. It helps a team choose where to review conditions before the event unfolds.

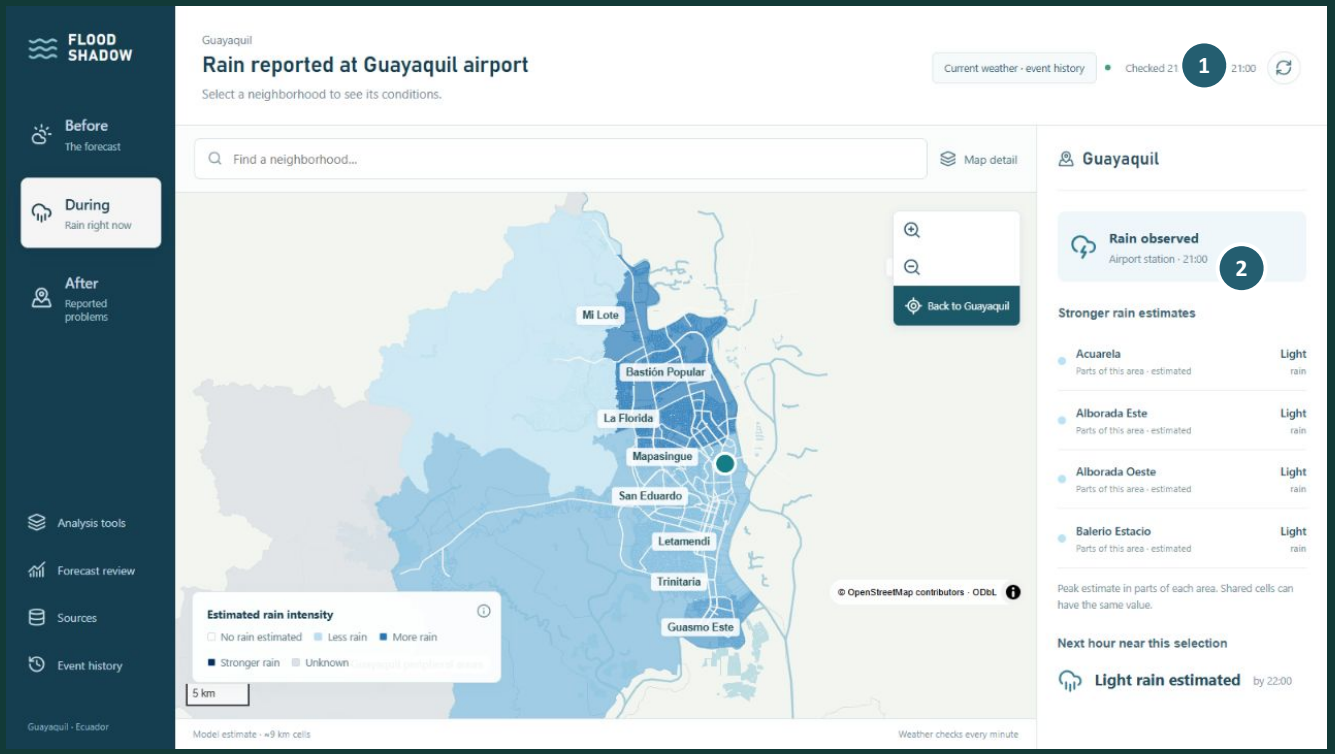
04 / DURING

Follow changing rain across Guayaquil

The city map updates with the latest available weather values. Area names, intensity colors and source times help the reader understand what changed.

AUTOMATIC CHECKS

The open view checks its weather connection every minute. A source can return the same value, so the display keeps the source time visible.



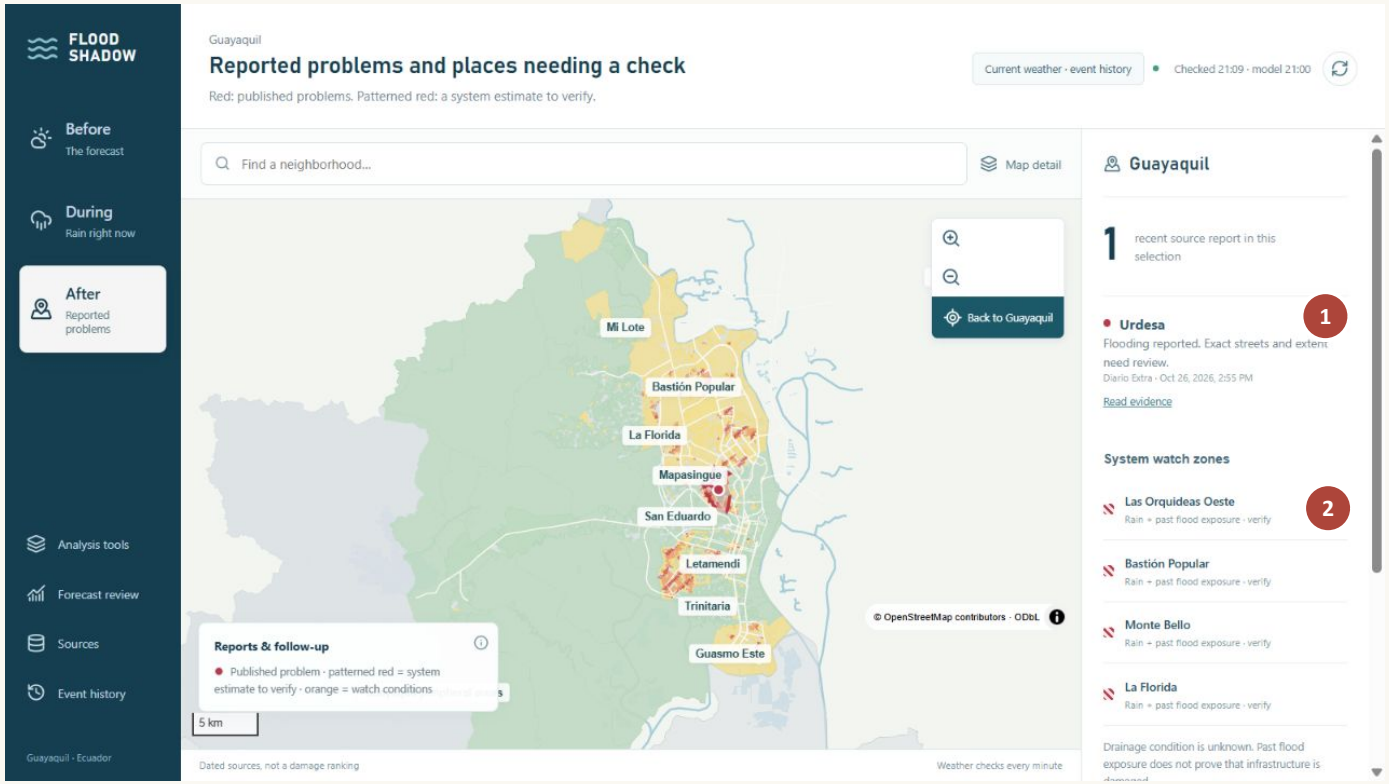
Real beta capture · Model estimates and a point observation remain distinguishable.

1 Source time
Check the model or station time, not only the time the page refreshed.

2 Complementary evidence
A city estimate and an airport observation describe different spatial views of the weather.

05 / AFTER

Document the impact and decide what to check next



Real beta capture · A verification priority is kept separate from confirmed or published damage.

1 Documented impacts

A published report provides evidence of a problem in an area. The reader can open the source and review its timing and scope.

2 Possible blind spots

Patterned zones identify places where rainfall and mapped exposure suggest that a localized check is useful, including where reports are sparse.

● **Solid red** Published problem

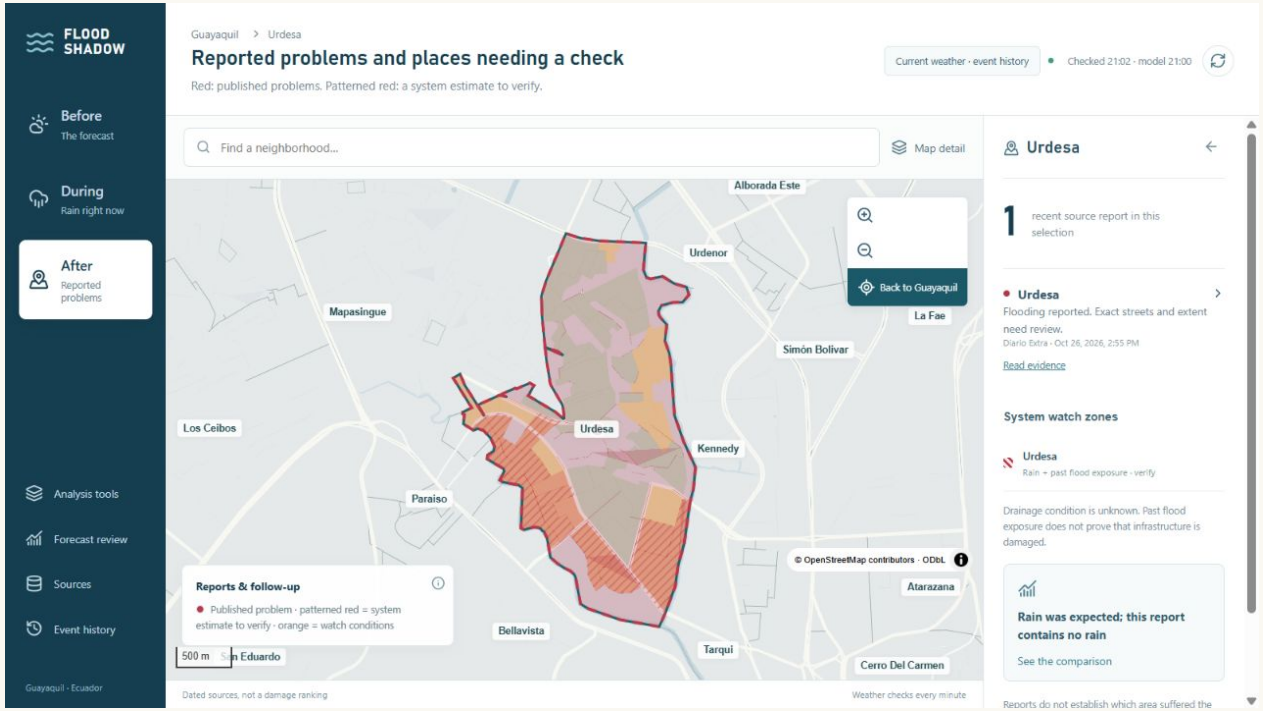
//// **Patterned red** System watch zone

The event record supports a later review of what was expected, what was reported and what still needs verification.

06 / INNOVATION

Make possible unreported impacts visible

FLOODSHADOW connects weather with territorial exposure and the availability of evidence. A place without a report remains a question to investigate.



Actual Urdesa selection · A published area reference and finer system watch polygons can be inspected together.

Territorial context

The city is organized into recognizable areas and official census units. Rain is interpreted in relation to that territory.

Reporting blind spots

The analysis gives visibility to exposed places that may not yet have a localized report. It supports a reasoned verification queue.

An event that remains useful

Saved outlooks and reviewed findings support comparison after the rain. The final product would use verified outcomes to improve future models.

Difference from a weather-only view: forecast + territory + impacts + a traceable next action.

07 / PRODUCT VISION

From the demonstrated flow to the final product

The beta demonstrates the connected experience. The final product would integrate richer infrastructure information, operational alerts and learning from verified events.

DEMONSTRATED IN THE BETA

Weather and map

Connected forecasts, estimated current rain, source dates and named-area selection.

Evidence and follow-up

Published impact references, system watch zones and preserved event or verification records.

Assessment review

Saved forecasts can be compared with compatible later evidence without rewriting the original outlook.

PROPOSED FINAL INTEGRATION

Infrastructure

Connect verified drainage, street access and essential-service data, including coverage and condition.

Operational alerts

Turn agreed watch criteria into traceable alerts with responsible recipients and follow-up status.

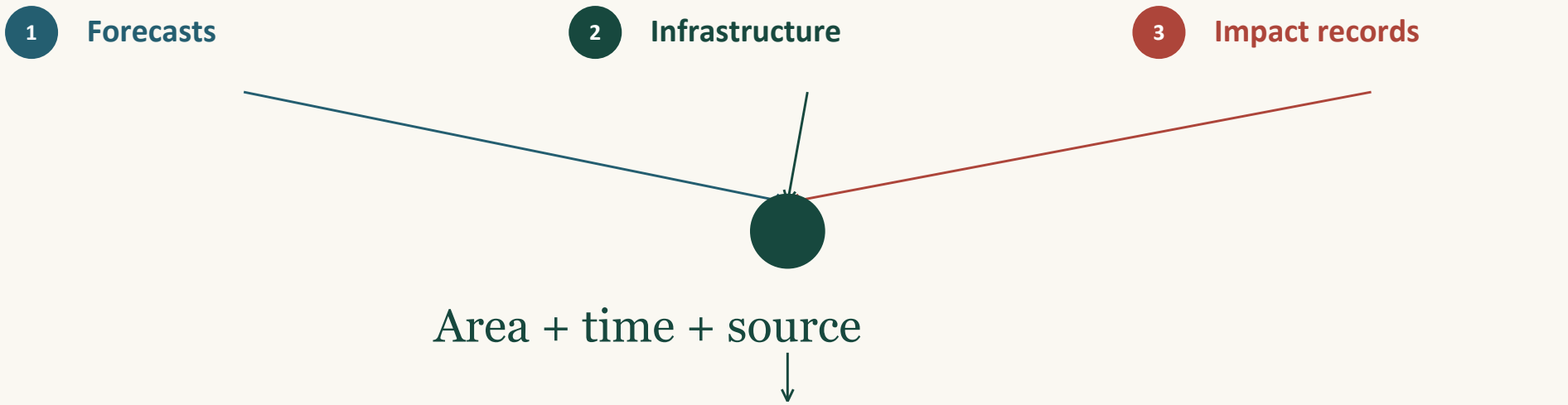
Machine learning

Train and evaluate models using independently verified outcomes from multiple events, then update future assessments.

Learning distinction: retaining evidence and updating an assessment already belong to the workflow. Automatic model retraining is part of the proposed final integration.

08 / INTEGRATION AND TOOLS

Connect information around a shared event



Final product loop: verified event outcomes improve the next assessment.

Tools in everyday terms: ECMWF through Open-Meteo supplies the weather outlook; the territorial model prioritizes attention. React, MapLibre, Node.js, Python tools and SQLite connect the interface, map and saved information.

09 / IMPACT AND IMPLEMENTATION

Useful decisions for Guayaquil

The expected impact is a more coordinated reading of the same storm, with clear reasons for preventive attention, monitoring and follow-up.

Before

Example: prioritize drainage or access checks in places combining exposure with a stronger rainfall outlook.

During

Example: send a local verification team to inspect an area whose estimated conditions intensified.

After

Example: document a reported access problem and retain the finding for later event review.

CURRENT IMPLEMENTATION EVIDENCE

66 / 66

Automated tests passed

6,652

Prepared census sectors

3 events

Preserved event datasets

IMPLEMENTATION PATH

Connect verified infrastructure → review with local users → collect representative checks → evaluate on independent events → deploy monitored alerts.

Impact evaluation: measure time to find evidence, correct map interpretation and useful findings per verification effort. These software checks do not establish rainfall precision, damage prevented or lives saved.

10 / SOURCES AND PROJECT INFORMATION

The sources behind the proposal

Open-Meteo

Weather outlook and ECMWF rainfall estimates.

open-meteo.com/en/docs

NOAA Aviation Weather Center

SEGU airport observations and report history.

aviationweather.gov/data/api

INAMHI

Daily guidance and Ecuador weather context.

servicios.inamhi.gob.ec/pronostico

INEC and Guayaquil nomenclature

Census geography and recognizable local area names.

Municipal sector nomenclature

INEC Geoportal

2022 census-sector layer.

INEC · Marco Geoestadístico 2022

Territorial susceptibility

Published Zonas Inundables GYE footprint.

ArcGIS · Zonas_Inundables_GYE

Diario Extra

Source-reviewed Urdesa event reference.

Guayaquil storm report · 26 October 2026

OpenStreetMap and MapLibre

Map attribution and geographic display.

OpenStreetMap contributors

Verification date: 26 October 2026. The current run passed 66 tests, the production build and prepared-data integrity checks. Screenshots show actual beta states and their source times; the urban drawing and product diagram are explanatory.

Development declaration: OpenAI Codex assisted implementation and documentation; Team 3Jota owns and explains the project decisions. **Submission scope:** English project proposal for the Ecuador challenge. The required English video pitch is a separate deliverable.

Visual credits: FLOODSHADOW captures and diagrams / 3Jota. Cover photograph: Freddy eduardo, *Vista del Malecón 2000*, Wikimedia Commons; cropped, CC BY-SA 4.0. Original image.